MILITARY SPECIFICATION

TANKER MOORING SYSTEM, MULTI-LEG

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a multi-leg tanker mooring system and pipeline terminal for installation offshore on the ocean floor.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
TT-V-121	- Varnish, Spar, Water-Resisting.
UU-T-81	- Tags, Shipping and Stock.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed, and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.

FSC 3835

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproof, Flexible.
MIL-C-3774	- Crates, Wood; Open 12,000- and 16,000-Pound Capacity.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-T-22085	- Tape, Adhesive, Preservation and Sealing.
MIL-B-22191	- Barrier Materials, Transparent, Flexible, Heat Sealable.
MIL-B-26195	- Boxes, Wood-cleated, Skidded, Load-bearing Base.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-D-52838	- Diving Equipment Set, Packaging of.
MIL-C-52950	- Crate, Wood, Open and Covered.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

TA13225E8219	- Set Tanker Mooring System Multi-Leg.

SUPPLY CATALOG

SC 3835-97-CL-E05	- Tanker Mooring System, Multi-Leg

(Copies of specifications, standards, drawings, and supply catalogs required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Att'n Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The tanker mooring system shall consist of the following functional parts: (1) Beach Launch and Recovery Equipment; (2) Mooring Site, Survey Equipment; (3) Mooring Leg, Development Equipment; (4) Pipe Terminal and Launching Equipment; and (5) Miscellaneous and Support Equipment.

3.2 First article (first produced tanker mooring system). The contractor shall furnish one or more tanker mooring systems as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce tanker mooring systems that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to the surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.5).

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3.4 Components. The sizes, types and quantities of components shall be as specified in Supply Catalog SC 3835-97-CL-E05 and TA13225E8219.

3.5 Government furnished property. When specified (see 6.2) the following property in the quantities indicated will be furnished by the Government (see 6.4):

Description	Identification	Quantity
Boat, Utility	(06101) Model MAR III MSB	2

3.6 Workmanship. All parts and components of the tanker mooring system shall be clean and free from extraneous material. Any necessary rework shall be limited to procedures which will not affect the strength or functioning of parts or components.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

(a) First produced tanker mooring system (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

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4.3 First produced tanker mooring system inspection.

4.3.1 Sampling. From each first produced tanker mooring system, a random sample of one each component of each functional part furnished by the contractor shall be selected for examination and test.

4.3.2 Examination. The first produced samples selected in accordance with 4.3.1 shall be examined as specified in 4.5.1. Presence of one or more defect shall be cause for rejection.

4.3.3 Tests. The first-produced samples selected in accordance with 4.3.1 shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. From each first produced tanker mooring system, a random sample of one each component of each functional part furnished by the contractor shall be selected for examination and test.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. Presence of one or more defect shall be cause for rejection.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The tanker mooring system shall be examined as specified herein for the following defects:

101. Material not as specified.
102. Components and parts missing or not as specified.
103. Workmanship not as specified.

4.5.2 Tests. Unless otherwise specified, one each component of each functional part shall be tested and accepted in accordance with the referenced component specification.

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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product . For the purpose of inspection, a complete pack (box, crate, bundle, unit load) as prepared for shipment shall be considered a unit of product.

4.6.1.2 Examination. Each unit of product shall be examined for the following defects. Presence of one or more defect shall be cause for rejection:

104. Materials, methods, and containers not as specified for Level A. Each incorrect material, method, or container shall be considered one defect.
105. Disassembly not as specified for Level A.
106. Matchmarking not as specified for Level A.
107. Components of the system procured by military or Federal specifications not preserved and packed in accordance with the component specification for Level A.
108. Components of similar design and construction not preserved and packed as specified for components procured by military or Federal specifications for Level A.
109. Unprotected exterior metal surfaces not coated with preservative as specified for Level A.
110. Coated machined surfaces not wrapped or covered with barrier material as specified for Level A.
111. Engines and engine accessories not preserved in accordance with the referenced document for Level A or Level C.
112. Cooling system not preserved as specified for Level A or Level C.
113. Utility boat not preserved as specified for Level A or Level C.
114. Diving equipment sets not preserved in accordance with the referenced document as specified for Level A.
115. Lumber not secured in bundles as specified for Level A.
116. Tools not preserved in accordance with the referenced document for Level A.
117. Components from different system (Parts I, II, III, IV, or V) packed together for Level A.
118. Blocking, bracing, and anchoring not in accordance with the referenced document for Level A.
119. Utility boat not contained in cradle as specified for Level A or Level C.
120. Strapping not as specified for Level A.
121. Marking missing, illegible, incorrect, or incomplete for Level A or Level C.

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5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Level C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. Disassembly or sectionalization of components procured by military or Federal specifications shall be in accordance with the Level A requirements of the component specification. Disassembly of all other components shall be in accordance with the contractor's recommendations, subject to the approval of the contracting officer, that will provide the minimum disassembly necessary to comply with the transportability requirements and the packing requirements specified in 5.2, and to safeguard parts vulnerable to damage, pilferage, and loss. Small components or sections that can be secured to the larger section or component from which they were removed shall be secured to that component or section by bolting, with soft annealed wire, or by strapping conforming to QQ-S-781, Class 1, Type I or IV, Finish B, size as applicable. Removed bolts, nuts, pins, and washers shall be reinstalled in mating parts and secured to prevent loss, or placed in a close-fitting box conforming to PPP-B-636, Class Weather-Resistant. The box shall be closed and sealed as specified for Method V in the appendix to the box specification and the box secured to the component or section for which intended.

5.1.1.2 Matchmarking. When necessary to facilitate reassembly, removed parts or assemblies and mating parts or assemblies shall be matchmarked. Matchmarking shall be by legible letters or numerals using lusterless white enamel overcoated with varnish conforming to TT-V-121. When practical because of size or configuration, small assemblies removed may be matchmarked and identified with tags conforming to UU-T-81, Type A. Information on the tags shall be applied with waterproof ink.

5.1.1.3 Components. Components of the system procured by military and Federal specifications shall be preserved in accordance with the Level A requirements of the component specification. Components of similar design and construction, procured by means other than military and Federal specifications, shall be preserved in a smiliar manner as those components procured by specification. Preservation of components for which no specification or guidance is furnished shall be provided protection adequate to prevent damage and deterioration against the most severe condition known or anticipated to be encountered during worldwide shipment, multiple handling, and indeterminate storage.

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The preservation shall be as determined by the contractor, subject to the approval of the contracting officer, utilizing the criteria established in MIL-P-116. The following minimum requirements will be adhered to.

5.1.1.3.1 Preservation. Preservatives specified herein shall conform to the applicable specification listed in, and shall be applied in accordance with, MIL-P-116.

5.1.1.3.2 Unprotected surfaces. Unprotected exterior metal surfaces of components requiring the application of a contact preservative in accordance with MIL-P-116, shall be preserved as follows:

5.1.1.3.2.1 Unfinished surfaces. Unfinished surfaces shall be coated with Type P-1 preservative.

5.1.1.3.2.2 Machined surfaces. Exposed exterior machined surfaces shall be coated with Type P-6 or P-11 preservative. The coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, Type I or II, Grade A, Class 2, or MIL-B-22191, Type II, and the material secured in place with tape conforming to MIL-T-22085, Type II.

5.1.1.3.3 Electronic and highly critical mechanical components. Electronic and highly critical mechanical components which require the highest degree of protection from the effects of water vapor shall be preserved in accordance with MIL-P-116, Method II, submethod as applicable.

5.1.1.3.4 Engines. Engines and engine accessories shall be preserved in accordance with MIL-E-10062, type as applicable, Method I, Level A.

5.1.1.3.4.1 Cooling system. Engines with liquid cooling systems shall be preserved by the water and antifreeze procedure, except the antifreeze shall conform to MIL-A-46153.

5.1.1.3.5 Utility boat. The engine and engine accessories shall be preserved in accordance with MIL-E-10062, Type III, Level A. The operating equipment and supplies furnished with the boat such as the tiller, anchor, and lifting sling, that require no preservation shall be stowed within the boat and secured. Equipment and supplies requiring a box to prevent pilferage, loss, deterioration, or damage shall be placed together in one or more boxes, as required, conforming to PPP-B-636, Class Weather-Resistant. The contents shall be cushioned and blocked within the box to prevent movement or damage. All hull drain openings shall be secured in the open position.

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5.1.1.3.6 Diving equipment sets. Diving equipment sets shall be preserved in accordance with the Level A requirements of MIL-D-52838.

5.1.1.3.7 Lumber. Lumber of like description shall be bundled together in quantities most convenient for handling and compatible with transportability requirements. Securing of the bundles shall be with strapping conforming to QQ-S-781, Class I, Type I or IV, Finish B, size as applicable.

5.1.1.3.8 Tools. Hand tools shall be preserved in accordance with the Level A preservation and packaging requirements of PPP-P-40.

5.1.2 Level C. Components procured by military or Federal specifications shall be preserved in accordance with the Level C or Commercial requirements of the component specification. Components procured by means other than military or Federal specification shall be preserved in accordance with the contractor's recommendations, subject to the approval by the contracting officer, utilizing the criteria established in MIL-STD-1188 with the following specific requirements:

5.1.2.1 Engines. Engines, including the utility boat engine, shall be preserved in accordance with the Level C alternate preservation and packaging requirements of MIL-E-10062, except cooling systems shall be preserved as specified in 5.1.1.3.4.1, and crankcases shall be filled with the operational lubricant.

5.1.2.2 Hull openings. All hull drain openings shall be secured in the open position.

5.2 Packing. Packing shall be Level A or Level C as specified (see 6.2).

5.2.1 General requirements. The Tanker Mooring System, Multi-Leg, consists of Parts I, II, III, IV, and V. Packing of components of one Part with components of another Part shall not be permitted.

5.2.1.1 Level A. Packing weights and dimensions shall comply with all applicable carrier rules and regulations, including those for air, to insure required transportability. When there is no conflict, components of the system procured by military or Federal specifications shall be packed in accordance with the Level A requirements of the component specification. Components of similar design and construction, procured by means other than military or Federal specification, shall be packed in a manner similar to those components procured by specification.

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Packing of components for which no specification or guidance is furnished shall be in accordance with the following requirements and as determined by the contractor, subject to approval of the contracting officer.

5.2.1.1.1 Boxes. Components not covered by commodity specifications requiring the protection afforded by a wood box to aid in the consolidation, handling, or to prevent damage, pilferage, or loss, shall be packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional, PPP-B-621, Class 2, style optional, or MIL-B-26195, Type II, style and class optional. The specified weight limitation of the containers shall not be exceeded. The contents shall be cushioned, blocked, braced, and anchored within the container to prevent movement or damage in accordance with MIL-STD-1186. Box closure and strapping shall be in accordance with the appendix to the applicable box specification. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, Finish B, size as applicable. Boxes that are to be consolidated in a crate (see 5.2.1.1.4) shall not require strapping.

5.2.1.1.2 Crates. Large components not covered by commodity specifications requiring the protection afforded by a wood crate to aid in consolidation, handling, or to prevent damage, pilferage, or loss, shall be packed in crates conforming to MIL-C-104, type class and style optional; MIL-C-3774, Type I or II; or MIL-C-52950, Style A, type as applicable. The contents shall be blocked, braced, and anchored to the crate base in a manner to obtain minimum cube, and to prevent movement or damage. Securing shall be in accordance with MIL-STD-1186 and the appendix to the crate specification. Strapping shall be as specified in 5.2.1.1.1.

5.2.1.1.3 Utility boat. The utility boat, preserved as specified in 5.1.1.3.5, shall be contained and secured in the cradle of the Boat Launch and Recovery Device Assembly. When the boat is procured separately, it shall be placed and secured in a cradle of the contractor's design. Equipment and supplies contained in fiberboard boxes shall be packed in wooden boxes specified in 5.2.1.1.1 and secured within the boat hull in a protected location.

5.2.1.1.4 Consolidation. When advantageous to aid in handling, shipping, storage, or inventory control, boxed or unboxed components may be packed together in crates as specified in 5.2.1.1.2.

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5.2.2 Level C. Components comprising the complete tanker mooring system shall be packed in accordance with the general requirements specified in 5.2.1 and in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules. The utility boat shall be contained in a cradle as specified in 5.2.1.1.3.

5.3 Preservation and Depreservation Guide. Two copies of the Depreservation Guide for Vehicles and Equipment, conforming to DA Form 2258, or Preservation and Depreservation Guide for Marine Equipment, conforming to DA Form 3256, as applicable, shall be prepared. One copy shall be placed in the package of technical publications and the package marked to indicate, "Preservation and Depreservation Guide Enclosed". The other copy shall be placed in a sealed waterproof envelope and marked as specified above, then secured in a conspicuous location on the commodity for which intended.

5.4 Marking. Marking for shipment and storage shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tanker mooring system is used to construct and install a multi-leg mooring system and pipeline terminal on the ocean floor to transfer petroleum products from tanker transports to onshore-tanks, pipelines or other storage facilities.

6.2 Ordering data. Procurement documents shall specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the first produced tanker mooring system and the number of mooring systems required (see 3.2).
(c) When Government-furnished property is required (see 3.5).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First produced tanker mooring system. Any changes or deviations of production tanker mooring systems during production will be subject to the approval of the contracting officer. Approval of the first produced tanker mooring system will not relieve the contractor of his obligation to furnish mooring systems conforming to this specification.

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6.4 Government-furnished property. The contracting officer shall arrange to furnish the property specified in 3.5, when applicable (see 6.2).

6.5 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

6.6 DA Form 2258 and DA Form 3256. The contracting officer should arrange to furnish DA Form 2258, Depreservation Guide for Vehicles and Equipment and DA Form 3256, Preservation and Depreservation Guide for Marine Equipment, when requested by the contractor.

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